

Uncertainty Quantification of Underwater Acoustic Transmission Loss

Monte Carlo · Delta/FOSM · PCE · LN3

Stochastic Bellhop Project

June 20, 2026

Motivation: Why Stochastic Transmission Loss?

Sonar planning requires TL, but inputs are uncertain:

- Source frequency f drifts with temperature
- Source depth z_S fluctuates with platform motion
- Sound Velocity Profile (SVP) varies with ocean state

Nominal deterministic TL (single Bellhop run) gives no information about **how much** TL can vary and **how likely** a target is detectable.

This project: compare four UQ methods for cost vs accuracy across 4 range/depth scenarios

Key Question

Given $\theta \sim \mathcal{N}(\mu, \sigma^2)$,
what is $P(\text{TL}(\theta) < \text{FOM})$?

(Shadow probability / detection probability)

Figure of Merit (FOM)

Detection threshold in dB.

$$P_{\text{detect}} = P(\text{TL} < \text{FOM})$$

Problem Setup: Scenarios, Parameters, Distributions

4 bathymetry scenarios:

- **deep_water** — isovelocity, flat bottom
- **shallow_water** — continental shelf
- **upslope** — rising bathymetry
- **downslope** — descending bathymetry

3 uncertain parameters (perturbed independently):

$$f \sim \mathcal{N}(\mu_f, (\frac{p}{200}\mu_f)^2)$$

$$z_c \sim \mathcal{N}(\mu_z, (\frac{p}{n} \mu_z)^2)$$

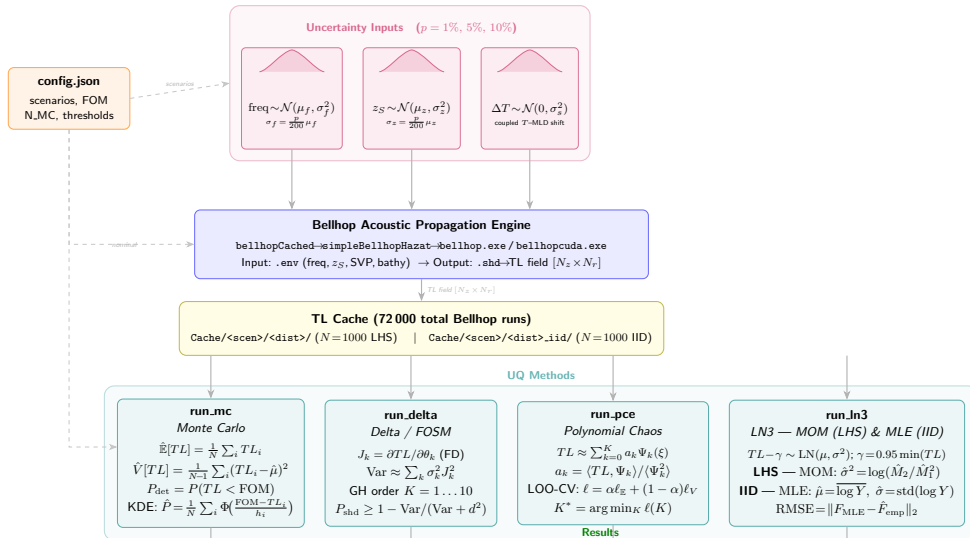
3 perturbation levels: $p \in \{1\%, 5\%, 10\%\}$

Combos	$4 \times 3 \times 3$	= 36 total
MC per combo	$N = 1000$	LHS samples
Total Bellhop runs	36 000	
Output grid	$N_z \times N_r$	depth $\times$ range

UQ methods compared:

- 1 Monte Carlo (MC) — reference
- 2 Delta / FOSM — 1st-order Taylor
- 3 PCE — Polynomial Chaos Expansion
- 4 LN3 — 3-parameter Lognormal MLE

Pipeline Overview



Method 1: Monte Carlo ($N = 1000$ LHS)

Estimators:

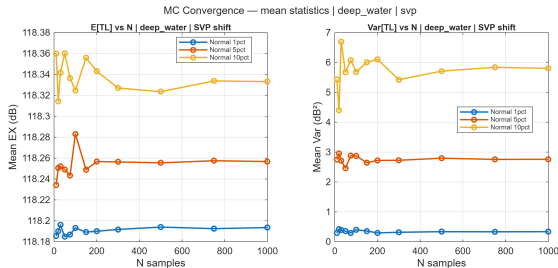
$$\hat{\mathbb{E}}[\text{TL}] = \frac{1}{N} \sum_{i=1}^N \text{TL}_i$$

$$\hat{\text{Var}}[\text{TL}] = \frac{1}{N-1} \sum_{i=1}^N (\text{TL}_i - \hat{\mu})^2$$

$$\hat{P}_{\text{det}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{TL}_i < \text{FOM}]$$

Sampling: Latin Hypercube Sampling (LHS) ensures stratified coverage of the input space with exactly $N = 1000$

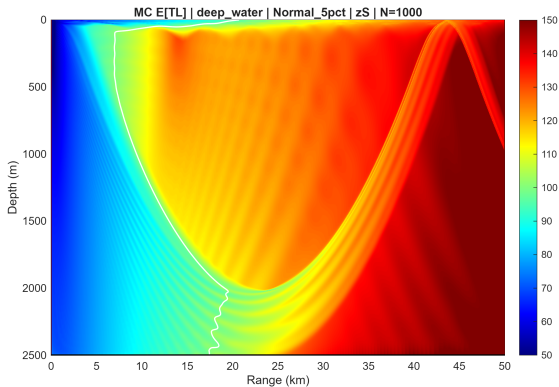
Convergence of $\hat{\mathbb{E}}$ and $\hat{\text{Var}}$ vs. N
(sub-samples of $N = 1000$ cache):



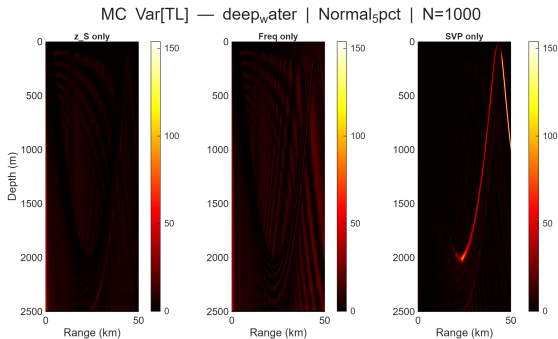
Mean converges by $N \approx 50$; Var requires $N \approx 200-500$.

MC Results: Mean TL and Variance Maps

Mean TL under z_S perturbation
(deep water, Normal 5%):



Variance maps for all 3 parameters:



Var[TL] is highest in the shadow zone transition region. Frequency perturbation dominates at long range.

Method 2: Delta / FOSM (First-Order Second-Moment)

1st-order Taylor expansion around nominal μ :

$$TL(\theta) \approx TL(\mu) + \sum_k J_k(\theta_k - \mu_k)$$

where $J_k = \partial TL / \partial \theta_k|_{\mu}$ (computed by finite differences — 3 extra Bellhop runs).

Variance propagation:

$$\text{Var}[TL] \approx \sum_k \sigma_k^2 J_k^2$$

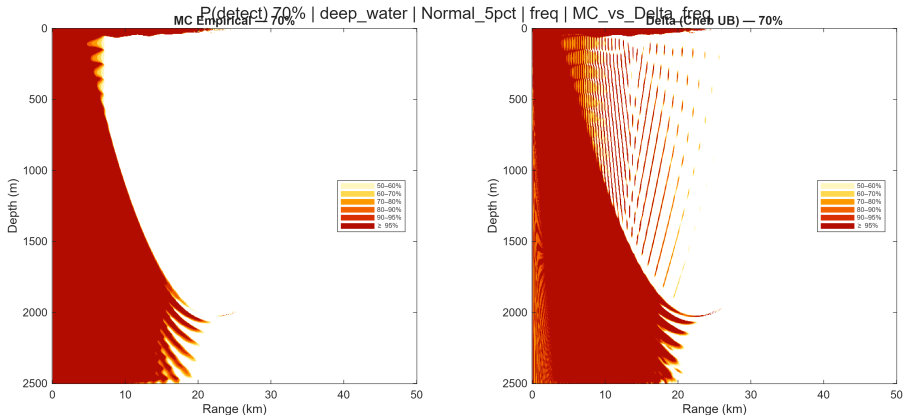
When Delta works

$\sigma \leq 1\%$, flat bathymetry — TL is nearly linear. Var maps accurate; shadow prob reliable.

When Delta fails

Slope environments, large σ — shadow zone boundary is non-smooth; linearisation breaks.

Delta Results: Shadow Probability Maps



Left: MC empirical P_{det} (reference). **Right:** Delta/Chebyshev P_{shd} (upper bound). At 1% perturbation the maps nearly coincide; discrepancy grows at $\geq 5\%$.

Method 3: Polynomial Chaos Expansion (PCE)

Approximate TL as a polynomial in the standardised input $\xi = (\theta - \mu)/\sigma$:

$$TL(\xi) \approx \sum_{k=0}^K a_k \Psi_k(\xi)$$

where Ψ_k are **Hermite polynomials** (orthogonal w.r.t. $\mathcal{N}(0, 1)$).

Coefficients via projection (reusing MC samples):

$$a_k = \frac{\langle TL, \Psi_k \rangle}{\langle \Psi_k^2 \rangle}$$

Then: $\mathbb{E}[TL] = a_0$,
 $\text{Var}[TL] = \sum_{k=1}^K a_k^2 \langle \Psi_k^2 \rangle$

LOO-CV Error (combined)

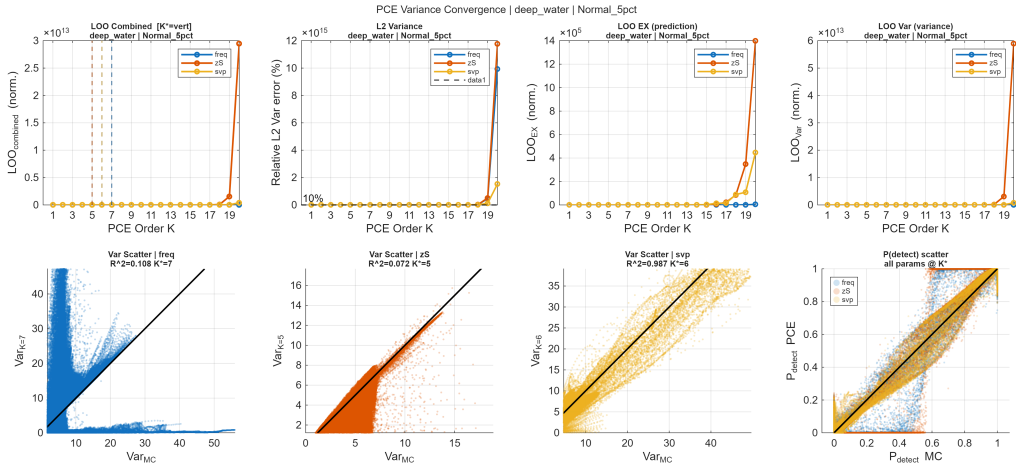
$$\ell(K) = \alpha \ell_{\mathbb{E}}(K) + (1 - \alpha) \ell_V(K)$$

$$K^* = \arg \min_K \ell(K)$$

Why not just maximise K ?

At $K \geq 18$, the Vandermonde matrix becomes ill-conditioned despite $N/(K + 1) \approx 47$. High-order Hermite polynomials blow up at $|\xi| > 2$, distorting the reconstructed

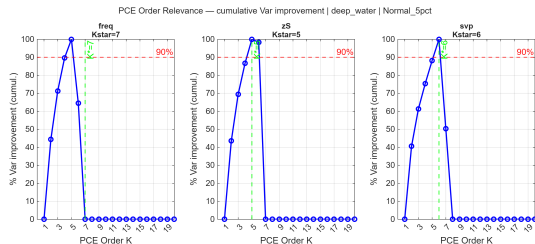
PCE: LOO-CV Order Selection (Deep Water, Normal 5%)



8 panel figure: top row — LOO-CV error vs. K for mean (blue), variance (orange), combined

PCE: Order Relevance and K^* Across All Combos

LOO error at K^* — how much each parameter “needs” higher-order expansion:



Selected K^* (all 12 scenario/dist combos):

Scenario / Dist	freq	z _S	SVP
deep / 1%	7	6	6
deep / 5%	7	5	6
deep / 10%	7	5	6
shallow / 1%	8	9	7
shallow / 5%	7	5	8
shallow / 10%	7	5	6
upslope / 1%	8	5	5
upslope / 5%	10	5	7
upslope / 10%	12	6	7
downslope / 1%	7	6	5
downslope / 5%	8	6	7

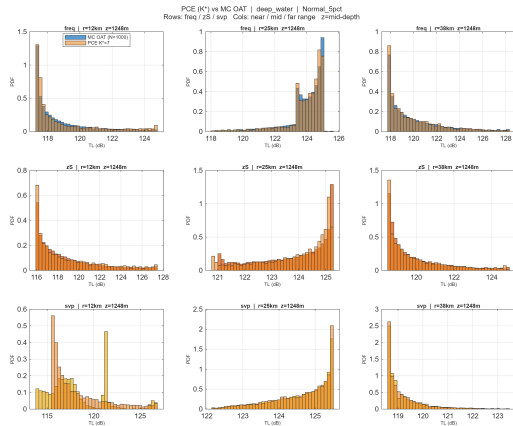
PCE vs MC: TL Distribution Comparison

Each pixel's TL distribution is compared between:

- **MC histogram** (1000 samples)
- **PCE(K^*) histogram** (synthetic samples drawn from the polynomial surrogate)

Key insight: Using $K = K_{\max} = 20$ instead of K^* causes **Hermite blow-up** at $|\xi| > 2$, producing a spurious PCE mean near 0 dB vs MC mean of 100–120 dB.

Using $K = K^*$ coefficients only gives correct distributions.



Method 4: 3-Parameter Lognormal (LN3) — MLE

Model: $TL - \gamma \sim \text{LN}(\mu, \sigma^2)$

Shift parameter: $\gamma = 0.95 \min(TL)$
(heuristic)

MLE estimation of (μ, σ) via MATLAB
`lognfit` on shifted samples $Y = TL - \gamma$:

$$\hat{\mu} = \overline{\log Y}$$

$$\hat{\sigma} = \text{std}(\log Y)$$

Shadow probability:

$$P(TL < \text{FOM}) = \Phi_{\text{LN}}(\text{FOM} - \gamma; \hat{\mu}, \hat{\sigma})$$

MLE vs MOM

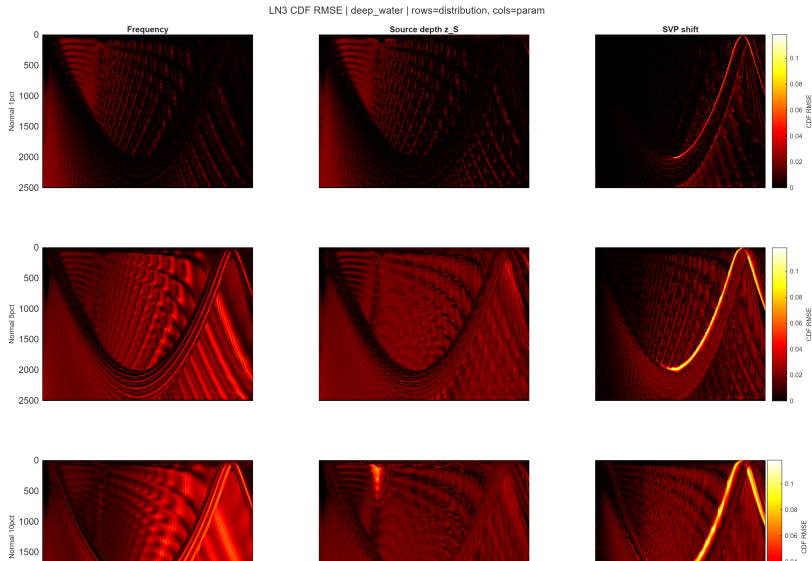
Method of moments (MOM)
estimator:

$$\hat{\sigma}_{\text{MOM}}^2 = \log(\hat{M}_2 / \hat{M}_1^2)$$

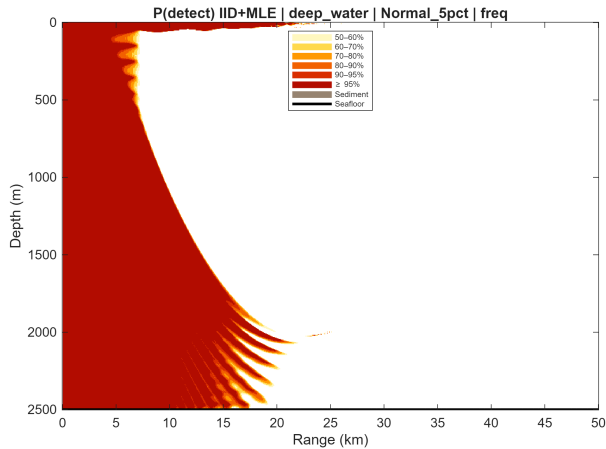
$$\hat{\mu}_{\text{MOM}} = \log(\hat{M}_1) - \frac{1}{2} \hat{\sigma}^2$$

MLE outperforms MOM at nearly all pixels (lower KS statistic), especially at small sample sizes and near-symmetric distributions.

LN3 Results: KS Goodness-of-Fit Heatmap



Detection Probability: Three Estimators Compared



Estimator	Type	Properties
Empirical MC	exact (ecdf)	gold standard sampling noise
MLE-LN3	param. fit	smooth maps fails bimodal
Chebyshev	upper bound	conservative always valid

Use **MLE-LN3** where $KS < 0.05$.

Use **Chebyshev** when only mean and variance are available (Delta method).

Use **Empirical MC** as ground truth.

Comprehensive Method Comparison

Method	Extra Cost	$\sigma \leq 1\%$	$\sigma = 5\text{--}10\%$	Slopes
Delta	3 FD runs	Excellent	Marginal	Fails
PCE(K^*)	0 (reuses MC)	Good	Good	Marginal
LN3-MLE	0 (reuses MC)	Good	Good	Good
MC ($N = 1000$)	1000 runs	Gold std.	Gold std.	Gold std.

Recommended hierarchy

- 1 Delta at $\sigma \leq 1\%$, flat bathy
- 2 PCE at $\sigma = 5\text{--}10\%$, flat bathy
- 3 MC for slopes or z_S uncertainty

LN3 role

Use LN3-MLE for **distributional shape** and compact shadow-probability maps.
Not a primary variance estimator.

Scenario Comparison: Deep vs Shallow vs Slope

Scenario	PCE K^* (freq)	Delta accuracy	Key characteristic
deep_water	7 (all dists)	Excellent at 1%	Classic ray convergence zone; Var largest near CZ boundary
shallow_water	7–8	Excellent at 1%	Mode interference dominant; z_S uncertainty hardest
upslope	8–12	Fails at 5%+	Bathymetry traps energy; shadow zone very sensitive
downslope	7–11	Fails at 5%+	Energy leaks into deep water; large TL variance at transition

SVP Perturbation: Physical Interpretation

The SVP perturbation model applies a uniform temperature shift $\Delta T \sim \mathcal{N}(0, \sigma_S^2)$ to the entire profile, coupled with a mixed-layer depth (MLD) change:

- Warmer $\Delta T > 0$: steeper gradient, stronger downward refraction
- Cooler $\Delta T < 0$: weaker gradient, surface duct collapses

This creates **non-linear** shifts in shadow zone boundaries — particularly sensitive in upslope/downslope scenarios.

SVP uncertainty effect

SVP perturbation at 10% causes the largest variance maps among the three parameters in deep water, because the convergence zone boundary is highly sensitive to sound speed gradient changes.

PCE for SVP

K^* for SVP: 5–8 (flat), consistent convergence. SVP does **not** show the z_S -type resistance that prevents convergence in deep water.

Key Findings

- 1 **Delta is near-exact at 1% perturbation** for flat bathymetry. Only 3 extra Bellhop runs needed. Fails at slope boundaries where TL is non-smooth in parameter space.
- 2 **PCE converges with $K^* = 5-9$** for flat scenarios (freq, SVP). Slope environments reach $K^* = 10-12$, signalling that polynomial surrogacy approaches its limit. z_5 uncertainty in deep water **resists PCE convergence** even at $K_{\max} = 20$: the shadow zone shifts non-linearly with source depth.
- 3 **LN3-MLE is reliable** ($KS < 0.05$) across most of the range-depth grid. Fails at bimodal interference nodes; use empirical MC or Chebyshev there.
- 4 **All analytical methods reuse the $N = 1000$ MC cache** at zero marginal Bellhop cost. The question is not “can we afford PCE?” but “**does the PCE surrogate add information beyond empirical MC?**” For flat bathymetry at

Conclusions and Future Work

Conclusions:

- Four UQ methods benchmarked across 36 scenario/distribution combos
- Delta is the cheapest and best for small perturbations
- PCE extends reliable UQ to moderate perturbations at zero extra cost
- LN3-MLE provides smooth, compact shadow-probability maps
- MC remains necessary for slope environments and z_S uncertainty
- Detection probability can be bounded

Future Work:

- **3D bathymetry** — extend from 2D range-depth to 3D
- **Range-dependent SVP** — spatially varying sound velocity
- **Multi-parameter PCE** — joint expansion over all 3 params (currently 1D per-parameter)
- **Bayesian updating** — use sonar returns to narrow parameter uncertainty in real-time
- **Operational interface** — integrate with sonar planning